DSAII- HW7

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Problem 1

Let $P(i,j)$ be the maximum probability to reach post$(i,j)$from $(1,1)$. The subproblem is getting to (n,n) based on the previous move, either $(i-1,j)$ or $(i,j-1)$. We then have to multiply probabilities since the path is a product of the edge probabilities along the path.
Let $r(i,j)$ be the the probability on the edge going right from (i,j) and let $d(i,j)$ be the probability on the edge going down from (i,j).
The Cases for the problem are:

$P(1,1) = 1$ Base case, The cat starts here

$P(1,j) = P(1,j-1) \cdot r(1,j-1)$ for $j > 1$, 1st row, can only arrive from the left.

$P(i,1) = P(i-1,1) \cdot d(i-1,1)$ for $i > 1$, 1st col, can only arrive from above

$P(i,j) = max(P(i-1,j) \cdot d(i-1,j), P(i,j-1) \cdot r(i,j-1))$ for $i,j > 1$ In this DP, the computation order for the table is row by row from left to right since we have information from $P(i-1,j)$ and $P(i,j-1)$. The Runtime is $O(n^2)$ since we have to fill in the table, each with 1 unit of work.

The Augmentation for the DP is to maintain a 2nd $n \times n$ table, that record the predecessor which achieved the max, so wither

$choice(i,j) =$ down if $P(i-1,j) \cdot d(i-1,j) \geq P(i,j-1) \cdot r(i,j-1)$

$choice(i,j) =$ right otherwise.
For the 1st row, $choice(1,j) =$ right for all $j > 1$. For the first column, $choice(i,j) =$ down for all $i > 1$.
To recover the path, we have to starts at (n,n) and trace backwards to $(1,1)$.

Problem 2

Part A

If there are only 2 people, and $f(1) = 1$, then $f(2) = 1/2$.
Since there are 2 people, and the 1st person choose a chair uniformly at random, there is a 50-50 chance they sit in the assigned seat. the 2nd person also has the same chance to get their assigned seat.

Part B
To build the recurrence we have to consider 3 cases:

Person 1 sits in their assigned chair, this has probability $1/n$. In this case all other people sit in the correct chairs so the probability that person n gets their seat is 1.

Person 1 sits in chair n with probability $1/n$. This means that everybody $2..n-1$ sits in their assigned chair, then when person n arrives, they sit in Person 1's chair. This a guaranteed failure so the probability is 0.

Person 1 sits in chair j, where $2 \leq j \leq n-1$, with probability $1/n$. This means everybody 2..j-1 sits in their assigned chair, then when person j arrives, they must pick another chair uniformly at random. Person j is analogous to another 'drunk person' so this is where this recurs, giving us $f(n-j+1)$. All together $f(n) = 1/n(1 + 0 + \sum_{j=2}^{n-1} f(n-j+1))$, but we can get rid of the 0 so
$f(n) = 1/n(1 + \sum_{j=2}^{n-1} f(n-j+1))$
If we let $k = n-j+1$, so j goes from 2 to n-1 and k from, n-1 down to 2
$f(n) = 1/n(1 + \sum_{k=2}^{n-1} f(k))$
Part C
From Part A, we can use the $f(2) = 1/2$ as a guess, so $f(n) = 1/2$ for all $n \geq 2$
BC: $f(2) = 1/2$ which is trivial from the problem.
IH: assuming $f(k) = 1/2$ for all $2 \leq k \leq n-1$, we can use this later in the IS.
IS: We need to show $f(n) = 1/2$
$f(n) = 1/n(1 + (n-2)/2) = 1/n \cdot n/2 = 1/2$
Thus, $f(n) = 1/2$ for all $n \geq 2$

Problem 3

Part A
Lex X be a indicator Random Variable where $X_i = 1$ means the node failed, and $X =_i 0$ means it passed. $E[X_i] = p$ for any i from$[1, n]$. Since we have n nodes, we can use LOE such that $X = X_1 + X_2 + ..X_n$ is the total number of failed nodes, thus $E[X] = np$
Part B
Markov's says $P(x \geq a) \leq E[x]/a$.
For this problem, $a = 1$ thus the Upper Bound for the Probability that at lease One Node Failing is $P(x \geq 1) \leq np/1 = np$
Part C
Let E_i be the event that the node i fails. Then $P(E_i) = p$ for all i from$[1, n]$.
By the Union Bound $P(\bigcup_{i=1}^{n} E_i) \leq \sum_{i=1}^{n} P(E_i) = np$
This is the Upper Bound on $P(E_i)$ which is the probability that at least One Node Fails. For the Probability that No Nodes Fail we take the complement, and apply the bound
$P(\text{No Fails}) = 1 - P(\bigcup_{i=1}^{n} E_i) \geq 1 - np$

Problem 4

Part A

Let X_i br an indicator Random Variable, where $X_i = 1$ is row i is bad and 0 otherwise. Each node survives has probability 1-p so the probability that all nodes in a row, $\sqrt{n}$, survive is $(1-p)^{\sqrt{n}}$
Thus $P(x_i = 1) = 1 - (1-p)^{\sqrt{n}}$
Let $X = X_1 + X_2 + ...X_{\sqrt{n}}$ be the total number of bad rows. By LOE
$E[X] = \sum_{i=1}^{\sqrt{n}} E[X_i] = \sqrt{n} \cdot (1 - (1-p)^{\sqrt{n})}$

Part B

If there are exactly $n/2$ bad nodes, that are chosen uniformly at random from all n nodes, We can Let X_i be an indicator random variable where $X_i = 1$ if row i is bad and 0 otherwise.
Since exactly $n/2$ fialed nodes are chosen uniformly at random, the probability that none fail in row i is
$P(X_i = 0) = \binom{n-\sqrt{n}}{n/2} / \binom{n}{n/2}$
Then we can take the complement so
$P(X_i = 1) = 1 - \binom{n-\sqrt{n}}{n/2} / \binom{n}{n/2}$

If we let $X = X_1 + X_2 + ..X_{\sqrt{n}}$ by LOE we get
$E[X] = \sum_{i=1}^{\sqrt{n}} E[X_i] = \sqrt{n} \cdot (1 - \binom{n-\sqrt{n}}{n/2} / \binom{n}{n/2}))$